

15 A sound wave propagates from left to right through a gas.

Fig. 15(a) shows the positions of some gas molecules at a particular instant of time.

Fig. 15(b) shows the variation with time t of the displacement s of one of these particles.

The distance between particles P and Q is 0.26 m.



Fig. 15(a)

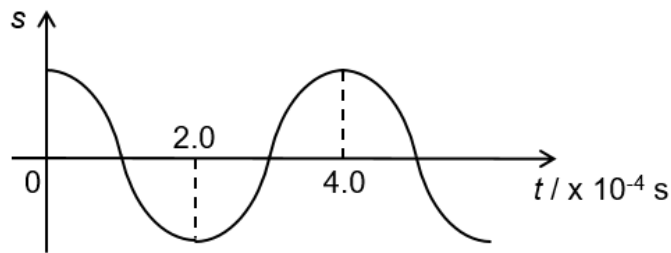


Fig. 15(b)

What is the speed of sound in this gas?

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|----------|-----------------------|----------|-----------------------|----------|-----------------------|----------|-----------------------|
| A | 300 m s ⁻¹ | B | 330 m s ⁻¹ | C | 380 m s ⁻¹ | D | 660 m s ⁻¹ |
|----------|-----------------------|----------|-----------------------|----------|-----------------------|----------|-----------------------|

Answer: B

$$\begin{aligned}
 v &= f\lambda \\
 &= \left(\frac{1}{T}\right)\lambda \\
 &= \left(\frac{1}{4.0 \times 10^{-4}}\right)\left(\frac{0.26}{2}\right) \\
 &= 325 = 330 \text{ m s}^{-1}
 \end{aligned}$$

